Review Questions

1. Differentiate between the following terms:

a. Data and Information

b. Bit and Byte

c. Pixel and Bitmap

2. Papa Eyram was asked by his daughter, Mawusi, to aid her in a computing homework. She was asked by her computing facilitator, Mr. Elikem, to convert the following decimal numbers into their binary representations. As a computing student, help Papa Eyram to also help his daughter.

a. 25

b. 57

c. 128

3. In a Science and Maths Quiz competition between three schools in the Upper West Region, the competing schools were to convert the following numbers into their decimal representations. Assuming you were one of the representatives of the schools, what would have been your answers to these questions, showing steps of working?

a. 11010

b. 100111

c. 1110100

4. If a computer has a 4-bit binary system, which of the following decimal numbers cannot be accurately represented, and why?

a. 7

b. 15

c. 16

d. 9

5. A digital speedometer of a moving vehicle shows 8 bits reading as 11001100. What is the speed of the vehicle in decimal value?

6. The binary representation of 9 is 1011. (True/False)

7. Match the decimal numbers with their binary representations:

- Decimal: 4
- Binary: ___
- Decimal: 17
- Binary: ___
- Decimal: 35
- Binary: ___

8. Sir Kojo was passing by to attend to one of his lessons and heard students in 3A General Arts class making noise. He then tasked the class to convert decimal number 22 into its binary representation. What would be your answer if you were a member of the class that day?

   A. 10101

   B. 11010

   C. 10110

   D. 11011

9. What does the AND operation do if both input values are 1?

   A. Outputs 0

   B. Outputs 1

   C. Outputs either 0 or 1 randomly

10. Which logic operation inverts the input value?

    A. AND

    B. OR

    C. NOT

11. The OR operation requires both inputs to be 1 to produce an output of 1. True or False?

    A. True

    B. False

12. Which of the following best describes the difference between RAM and ROM?

    A. RAM is volatile and ROM is non-volatile.

    B. RAM is used for permanent storage and ROM is used for temporary storage.

    C. RAM is non-volatile and ROM is volatile.

13. Given that 1 KB equals 1024 bytes, how many bytes are there in 7 KB?

14. Why is ROM considered non-volatile?

    A. It retains data without power.

    B. It loses data when power is turned off.

    C. It can be easily rewritten.

    D. It operates at high speeds.

15. Given two binary numbers A = 01101 and B = 10011, after performing the AND operation point wise (bit by bit) on A and B, how many 1s are in the output?

    A. 1

    B. 2

    C. 3

16. If a computer has 8GB of storage and a programme takes up 8000 MB of memory, can the programme be stored? Explain your reasoning.
17. A student claims that deleting files from a computer's ROM can speed it up. Evaluate this statement and explain why it is correct or incorrect.
18. Create a truth table for the following statement and evaluate when it is true:
    'Kojo is not an artist and Abena is not a musician'
19. What does CPU stand for?
20. Which part of the human body is the best analogy for what the CPU is to a computer?
    A. Heart
    B. Brain
    C. Eyes
    D. Skin
21. Which component is primarily responsible for executing instructions in a computer?
    A. RAM
    B. ROM
    C. CPU
    D. SSD
22. Identify the three main components of a CPU.
    A. Arithmetic Logic Unit (ALU), Control Unit (CU), and Registers
    B. Hard Drive, Monitor, and Keyboard
    C. Motherboard, Power Supply, and Fan
    D. RAM, ROM, and SSD
23. Explain the role of CPU registers; specifically, how they help make computers more efficient.
24. Explain the role of the ALU.
25. Explain the role of the CU.
26. How does increasing the clock speed of a CPU affect its performance?
27. If Efo Danny's personal computer's CPU can process 10 thousand instructions per second, how many instructions can it process in one minute?
28. Describe how cache memory benefits CPU performance.
29. What are the four stages of the machine cycle?
30. Name the three buses that make up the system bus.

31. What is the name of the set of commands that a processor can understand and execute?
32. Describe the steps involved in the fetch phase in the Fetch-Decode-Execute-Store cycle.
33. Research and prepare a presentation to illustrate, with examples, how a typical instruction is processed through the Fetch-Decode-Execute-Store cycle.
34. Explain the role the CPU clock plays in the machine cycle.

Answers to Review Questions

1. a. Data is the raw bits which is meaningless while information is the processed data which has meaning.

   b. A bit is the smallest unit of data and byte is eight contiguous bits.

   c. A pixel is the smallest unit of a digital image or display while bitmap is an array/grid of binary data representing the colour values of pixels in an image or display.`

2. 25 in binary is 11001 (Refer to pages 3 and 4 for steps to convert Decimal to Binary).

   57 in binary is 111001.

   128 in binary is 10000000.

3. 11010 in decimal is 26.

   100111 in decimal is 39.

   1110100 in decimal is 116.

4. In a 4-bit binary system, the highest number it can represent is 15 (1111 in binary). Therefore, the decimal number 16 cannot be accurately represented. 16 exceeds the maximum value a 4-bit system can represent, while 7, 15, and 9 can be accurately represented within a 4-bit limit.

5. The digital speedometer reading of 11001100 in binary corresponds to a speed of 204 in decimal.

6. False, 1011 represents 11.

7. - Decimal: 4, Binary: 100

   - Decimal: 17, Binary: 10001

   - Decimal: 35, Binary: 100011

8. C

9. B: output 1

10. C: NOT

11. B: False

12. A

13. Answer: 7KB = 7 * 1024 = 7168 bytes

14. B: it retains data without power

15. A:. 1

16. Yes, the programme can be stored. 8 GB is equal to 8192MB (since 1GB = 1024MB), which is more than the 8000MB required by the programme.

17. Incorrect. ROM (Read-Only Memory) is non-volatile and used to store firmware or permanent software to boot the computer. It is not used for general file storage, and users typically cannot modify or delete its contents.

18. Truth table:

    A = Kojo is an artist

    B = Abena is a musician

A	B	NOT(A)	NOT(B)	NOT(A) AND NOT(B)
1	1	0	0	0
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

    The statement is true only in case 4.

19. CPU stands for Central Processing Unit.
20. B: Brain
21. C: CPU
22. A: Arithmetic Logic Unit (ALU), Control Unit (CU), and Registers
23. CPU registers act as small, fast storage spaces within the computer's brain (the CPU). They store data, programme instructions and memory addresses. This quick access to data helps computers work more efficiently by allowing the CPU to store and retrieve information rapidly, enhancing the speed and performance of tasks like running programmes and performing calculations.
24. The Arithmetic Logic Unit (ALU) helps the CPU to do calculations and solve problems. For example, when you add numbers on a computer or compare two values, the ALU is the part of the CPU that does these tasks.
25. The CU interprets programme instructions and directs the flow of data between the CPU, memory, and other input/output devices, ensuring that tasks are executed in the correct sequence and according to the instructions provided.
26. Increasing the clock speed of a CPU improves its performance by allowing it to execute more instructions per second, leading to faster processing and better overall system performance.
27. The CPU can process 600,000 instructions in one minute.
28. Cache memory benefits CPU performance by providing a small, but extremely fast, memory area that stores copies of the data and instructions that are frequently accessed from the main memory (RAM).
29. Fetch, Decode, Execute and Storage.
30. Control bus, Address bus and Data bus.

31. Instruction Set or Instruction Set Architecture (ISA).

32. Steps involved in the fetch phase in the Fetch-Decode-Execute-Store cycle.

  a. The phase begins with the address of the instruction to be fetched stored by the Program Counter (PC).

  b. The address bus collects this address from the PC (which is then incremented to store the next location) and informs the CPU of the location of the instruction to be fetched.

  c. The data bus goes to this location and brings the instruction back to the CPU.

33. The presentation should include examples and visuals to explain each phase clearly and how they connect to form a complete cycle of instruction processing.

34. The clock generates regular electronic pulses that synchronise the various stages of the cycle, ensuring each step happens at the right time.

EXTENDED READING

- Click on the link below to learn more on ASCII Code and Binary https://youtu.be/H4l42nbYmrU?si=BxrlW76QNrAbi6zS
- Click on the link below to watch a video on binary numbers https://www.youtube.com/watch?v=kcTwu6TFZ08&t=2s
- Click on the link below to play a binary game https://learningcontent.cisco.com/games/binary/index.html
- CSC 170 – Introduction to Computers and their Applications – Lecture #1 – Digital Basics

Computational Fairy Tales:

- Pixels, Peacocks, and the Governor's Turtle Fountain https://computationaltales.blogspot.com/2011/06/pixels-peacocks-and-governors-turtle.html
- Using Binary to Warn of Snow Beasts https://computationaltales.blogspot.com/2012/01/using-binary-to-warn-of-snow-beasts.html
- Unhappy Magic Flowers and Binary https://computationaltales.blogspot.com/2011/06/unhappy-magic-flowers-and-binary.html
- Introduction to Boolean logic https://youtu.be/-hhZIFEGIIU
- Difference between SRAM and DRAM https://youtu.be/n974TtK0qnA
- What is a Browser Cache? How Do I Clear It? (youtube.com)
- CPU Cache Explained - What is Cache Memory? - YouTube

Computational Fairy Tales:

- Computer Memory and Making Dinner https://computationaltales.blogspot.com/2011/03/computer-memory-and-making-dinner.html
- Caching and the Librarian of Alexandria https://computationaltales.blogspot.com/2011/03/caching-and-librarian-of-alexandria.html

Watch this video on how a CPU works:

- https://www.youtube.com/watch?v=cNN_tTXABUA
- The Fetch-Execute Cycle: What›s Your Computer Actually Doing? - YouTube (9 mins)